OLA Booking Analysis Report using CSV Dataset, SQL Queries, and Dashboard

1. Introduction

This project was carried out to analyze ride-booking operations using an OLA booking dataset in CSV format. The dataset was imported into SQL for querying and then visualized in a dashboard tool to generate business insights. The main objective of this project was to understand booking trends, cancellation patterns, customer behavior, revenue generation, vehicle performance, and service quality.

The analysis was performed in three stages:

Dataset preparation using CSV

Data analysis using SQL queries

Dashboard creation for visual representation

The SQL logic used in the project included queries for booking status, revenue, vehicle performance, payment methods, cancellations, ride distance, and customer activity.

2. Objective of the Project

The key objectives of this project were:

- To analyze total ride bookings and their status
- To identify the most used vehicle types and locations
- To calculate revenue from successful bookings
- To study cancellation reasons and incomplete rides
- To evaluate customer and driver ratings
- To build dashboards for easy understanding of business performance

3. Dataset Overview

The project used a CSV dataset containing OLA ride booking records. The dataset included 10,200 rows and 19 columns, which makes it suitable for analysis and dashboard building.

Some important columns in the dataset were:

- Date – booking date
- Time – booking time
- Booking_ID – unique booking identifier
- Booking_Status – success, cancelled, incomplete

- Customer_ID – customer identifier
- Vehicle_Type – type of ride booked
- Pickup_Location and Drop_Location – ride locations
- V_TAT – vehicle arrival time
- C_TAT – customer turnaround time
- Canceled_By – who cancelled the ride
- Canceled_Reason – reason for cancellation
- Booking_Value – ride fare value
- Payment_Method – payment type
- Ride_Distance – distance of ride
- Driver_Ratings and Customer_Rating – service feedback

The data covered bookings from January 1, 2026 to March 2, 2026.

4. Tools and Technologies Used

The following tools were used in this project:

CSV File – source of raw dataset

SQL – for data querying and analysis

Power BI / Dashboard Tool – for creating interactive dashboards

Spreadsheet/CSV processing – for initial data review and cleaning

5. SQL Analysis Performed

The SQL queries used in this project focused on practical business questions such as booking distribution, revenue, vehicle performance, and customer activity. The query file included analysis such as total successful vs cancelled bookings, average booking value by vehicle type, top pickup locations, total revenue in the last 30 days, long-distance successful rides, preferred payment methods for high-value rides, ratings by vehicle type, and cancellation analysis.

Key SQL Queries Used

- Total number of bookings by booking status
- Highest average booking value by vehicle type
- Top 5 pickup locations by ride requests
- Revenue generated from successful rides in the last 30 days
- Successful rides above 20 km
- Most used payment method for rides above ₹500
- Average customer rating by vehicle type
- Driver-side cancellations due to refusal

- Monthly booking frequency by customer
- Average vehicle arrival time across locations

6. Major Findings from the Analysis

6.1 Booking Status Analysis

The dataset showed three main booking outcomes:

Successful bookings: 7,722

Cancelled bookings: 1,999

Incomplete rides: 479

This shows that the majority of rides were completed successfully, but cancellations and incomplete rides still formed a meaningful portion of operations.

6.2 Vehicle Type with Highest Average Booking Value

Among all vehicle categories, Lux had the highest average booking value, around ₹509.71. This indicates that premium rides generate higher revenue per booking compared to other vehicle types.

6.3 Top Pickup Locations

The most frequent ride request locations were:

HSR Layout – 1,523 rides

Airport – 1,505 rides

Koramangala – 1,471 rides

Indiranagar – 1,467 rides

Whitefield – 1,433 rides

These locations can be considered high-demand areas and may require better driver allocation and surge planning.

6.4 Revenue Analysis

The total revenue generated from successful bookings in the last 30 days of the dataset was approximately:

₹25,33,243.30

This metric is important for measuring recent business performance.

6.5 Long-Distance Successful Rides

There were 6,266 successful rides where the distance was greater than 20 km. This indicates that a significant share of bookings came from medium- to long-distance travel.

6.6 Preferred Payment Method for High-Value Rides

For rides with booking value greater than or equal to ₹500, the most frequently used payment method was:

Cash – 1,256 rides

This suggests that even for higher-value rides, a large number of customers still preferred cash payment.

6.7 Customer Ratings by Vehicle Type

The highest average customer rating was observed for:

Lux – approximately 3.10

This means customers were relatively more satisfied with premium ride experiences.

6.8 Driver Cancellation Analysis

The number of rides cancelled by drivers due to “Driver refused” was: 230 rides

This is an important operational issue because it affects customer satisfaction and booking reliability.

6.9 Frequent Customer Analysis

The original SQL requirement was to identify customers booking more than 10 rides in a month, but the dataset did not contain any such customer. Because of this, the threshold was adjusted in the SQL file to more than 5 rides for meaningful output.

This shows that the dataset mainly represents moderate-frequency users rather than very heavy monthly users.

6.10 Vehicle Arrival Time by Location

Average vehicle arrival time differed across pickup locations. The highest average V_TAT was seen in

HSR Layout – around 6.33

Airport – around 6.33

This may indicate traffic congestion or vehicle availability issues in these areas.

7. Dashboard Development

After the SQL analysis, dashboards were created to visually represent the performance of the ride-booking business. The dashboard helped convert raw numbers into understandable charts and KPIs.

Dashboard Elements Included

The dashboard can include the following visuals:

- Cards / KPIs
- Total Bookings
- Successful Bookings
- Cancelled Bookings
- Total Revenue
- Average Booking Value
- Bar Charts
- Bookings by Vehicle Type
- Top Pickup Locations
- Payment Method Usage
- Pie / Donut Charts
- Booking Status Distribution
- Cancellation Distribution
- Line Chart
- Revenue Trend by Date
- Table / Matrix
- Customer-wise or location-wise ride summary
- Rating Analysis
- Average customer rating by vehicle type
- Driver rating trends
- Benefits of the Dashboard

The dashboard made the analysis more interactive and easier to understand. It helped in:

1. Quickly identifying business trends
2. Comparing vehicle performance
3. Monitoring cancellation issues
4. Understanding customer behavior
5. Supporting data-driven decisions

8. Business Insights

From this project, the following business insights were obtained:

1. Successful rides form the major share, which shows stable service operations.

2. Cancelled and incomplete rides still need attention, as they directly affect customer trust.
3. Lux rides bring higher revenue and better ratings, showing stronger value in premium services.
4. HSR Layout, Airport, Koramangala, Indiranagar, and Whitefield are high-demand areas, so resource planning should focus there.
5. Cash is still heavily used, even for higher-value rides.
6. Driver refusal is an important cancellation reason, suggesting the need for better driver discipline and ride acceptance policies.
7. Long-distance rides form a large portion of successful trips, which can help in pricing and profitability planning.
8. Arrival time varies by location, indicating possible gaps in fleet distribution.

9. Challenges Faced

Some practical challenges were also observed in the project:

The dataset did not contain customers with more than 10 rides per month, so the query condition had to be adjusted.

The dataset did not include the exact cancellation reason “Personal Reasons,” so an available reason, Driver refused, was used instead in the SQL logic.

Data cleaning and validation were necessary before dashboard creation.

Some business questions required slight adaptation based on actual dataset values.

10. Conclusion

This project successfully demonstrated how a CSV dataset can be transformed into useful business insights using SQL queries and dashboards. The CSV file served as the base data source, SQL was used for analytical processing, and the dashboard was used for visualization and reporting.

The project helped in understanding operational performance, customer behavior, revenue trends, and service quality in a ride-booking platform. It also showed the importance of combining data storage, SQL analysis, and dashboard visualization in a complete analytics workflow.

Overall, this project is a good example of how real-world business data can be used to make better decisions through data analysis and reporting.

11. References

SQL query file used for analysis of OLA bookings, including booking status, revenue, ratings, cancellations, customer ride frequency, and arrival time analysis.